SNFS — Special Number Field Sieve

History, Hardware Requirements, and Feasibility for M_{1229}

Target: C284, the 284-digit composite cofactor of $M_{1229} = 2^{1229} - 1$
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Executive Summary

SNFS was invented in 1989–1990 by Arjen K. Lenstra, Hendrik W. Lenstra Jr., Mark S. Manasse, and John M. Pollard. The first major success was the factorization of the ninth Fermat number $F_9 = 2^{512} + 1$ in June 1990. The general NFS (GNFS) was actually derived from SNFS — SNFS came first.

Breaking news (September 2026): The current SNFS world record is now $M_{1277} = 2^{1277} - 1$ at SNFS difficulty 384 digits, factored just last month. This means M_{1229} (SNFS-370) is now EASIER than the standing record and has essentially already been "beaten" by the state of the art.

Bottom line: Factoring C284 of M_{1229} is a routine Cunningham-Project-scale SNFS job — harder than NFS@Home's current SNFS-340 work, but easier than the just-set M_{1277} record. A well-organized effort could complete it in ~1 year at ~$0 cash cost using volunteer sieving (NFS@Home) plus an HPC linear-algebra allocation (XSEDE/TACC).

1. History of SNFS

1.1 Inventors and Original Papers

John M. Pollard first proposed the Number Field Sieve in an unpublished 1988 manuscript, later reprinted in The Development of the Number Field Sieve (LNM 1554, Springer 1993). Around 1989–1990, Arjen K. Lenstra, Hendrik W. Lenstra Jr., Mark S. Manasse, and Pollard developed the Special Number Field Sieve — the version for numbers of the form $a \cdot b^n + c$ with small a, b, c. As Wikipedia's SNFS article notes: "The general number field sieve (GNFS) was derived from it" — i.e. SNFS came first, GNFS was the later generalization.

1.2 Major Milestones

Date	Number Factored	SNFS Difficulty	Team / Hardware
Jun 1990	F9 = 2^512 + 1	~155 digits	Lenstra[2] + Manasse + Pollard (workstations)
2005–06	6^353 − 1	275 digits	Aoki, Kida, Shimoyama, Ueda
2006–07	M1039 = 2^1039 − 1	313 digits (kilobit)	Aoki, Franke, Kleinjung, Lenstra, Osvik
Aug 2012	M1061 = 2^1061 − 1	320 digits	Childers + NFS@Home + Trestles/Lonestar (~335 CPU-yr)
2010–14	Mersenne Factory: 17 numbers, n=1061–1199	326–361 digits	Kleinjung, Bos, Lenstra (~7,500 CPU-yr on 2.2 GHz Opterons)
Dec 2014	M1199 = 2^1199 − 1	361 digits	Final Mersenne Factory result
Sep 2026	M1277 = 2^1277 − 1	384 digits (CURRENT RECORD)	Recent; reported Wikipedia int. fact. records

1.3 Key Researchers

Name	Affiliation	Contribution
John M. Pollard	UK, independent	Inventor of NFS (1988), also p−1, rho
Arjen K. Lenstra	EPFL, Switzerland	Co-developer of SNFS; Mersenne Factory leader
Hendrik W. Lenstra Jr.	Leiden, Netherlands	Co-developer of SNFS; ECM inventor (1987)
Carl Pomerance	Dartmouth / Bell Labs	"A Tale of Two Sieves" survey; QS/NFS history
Peter L. Montgomery	(deceased Feb 2020)	Montgomery multiplication; central to all NFS codes

Name	Affiliation	Contribution
Thorsten Kleinjung	EPFL / Bonn	Polynomial selection, lattice sieving; MFF lead
Joppe W. Bos	NXP / EPFL	Mersenne Factory co-author
Emmanuel Thomé, Pierrick Gaudry, Paul Zimmermann	INRIA / LORIA (France)	CADO-NFS; RSA-240/250/260; GMP-ECM
Jason Papadopoulos	independent	Msieve (fast final processing)
Greg Childers	Cal State Fullerton	NFS@Home (BOINC sieving grid); M1061
Sam Wagstaff	Purdue / CERIAS	Cunningham Project maintainer
Ryan Propper	independent	ECM 83-digit record (2013, still standing)

2. Hardware Requirements

2.1 Calibration Data Points (Real Measurements)

The following table lists measured computation costs from real-world factorizations, providing the empirical basis for our C284 estimates.

Factorization	Type	Size	Effort	Hardware
M1061 (2012)	SNFS-320	320 digits	~335 CPU core-yr	NFS@Home + Trestles/Lonestar
Mersenne Factory (2014)	SNFS-326→361	17 numbers	~7,500 CPU-yr	2.2 GHz AMD Opterons
RSA-240 (Dec 2019)	GNFS-240	240 digits	~1,000 CPU-yr	World-wide grid
RSA-250 (Feb 2020)	GNFS-250	250 digits	~2,700 CPU-yr (2,450 sieve + 250 LA)	Xeon Gold 6130 grid
RSA-260 (Sept 2026)	GNFS-260	260 digits	~13.5 GPU-yr (~$400k)	Nvidia B200/GB200/GB300 (Cognition)
M1277 (Sept 2026)	SNFS-384	384 digits	CURRENT RECORD	Recently announced

2.2 SNFS vs GNFS Complexity (Theory)

The asymptotic complexity formulas are:

$$\text{SNFS: } L_n[1/3, (32/9)^{1/3}] = L_n[1/3, 1.526]$$

$$\text{GNFS: } L_n[1/3, (64/9)^{1/3}] = L_n[1/3, 1.923]$$

The work ratio between GNFS and SNFS grows rapidly with input size. For 284-digit inputs (the size of C284), the raw L-function work ratio is approximately 588× in favor of SNFS. In realized CPU-years the advantage is smaller (around 25–45×) due to differing constants, but the direction is overwhelming: SNFS is the only viable path for M_{1229}.

Decimal digits d	GNFS / SNFS work ratio (same d)
200	~29,000×
250	~83,000×
284 (C284)	~160,000×
320	~302,000×
370 (M1229 SNFS difficulty)	~686,000×
385	~866,000×

2.3 Estimated Hardware for SNFS at Various Difficulties

Using Mersenne Factory (M1199, SNFS-361, ~880 CPU-yr per number without sharing) as the most relevant 2014-era calibration point, scaling by $L_n[1/3, 1.526]$:

SNFS difficulty	Sieving (CPU-yr)	LA (CPU-yr)	Total (CPU-yr)	RAM for LA	Disk
200 digits	<1	<0.1	~1	16–64 GB / 1 node	~50 GB
250 digits	~2	~0.3	~2–3	64–256 GB / few nodes	~100 GB
280 digits	~22	~3	~25	0.5–2 TB / small cluster	~500 GB
300 digits	~88	~12	~100	1–4 TB / 8–16 nodes	~1–2 TB
320 digits (M1061)	~300	~35	~335	2–8 TB / 16–32 nodes	~2–5 TB
350 digits	~450	~70	~520 (shared)	4–16 TB / 16–64 nodes	~5–10 TB
370 digits (M1229)	~1,000–1,500	~150–400	~1,500–2,500	8–32 TB / 32–64 nodes × 256–512 GB	~5–15 TB
384 digits (M1277 record)	~2,400	~400–800	~3,700	16–64 TB / large cluster	~10–30 TB
400 digits	~6,000	~1,000	~7,000+	very large cluster	~30+ TB

2.4 Specific Estimates for M1229 / C284

$M_{1229} = 2^{1229} - 1$ has SNFS difficulty ≈ 370 digits (1229 bits). The cofactor C284 (284 digits) is what we want to factor, but sieving + linear algebra are sized by the 370-digit SNFS difficulty.

Sieving:

- ~1,000–1,500 CPU core-years, or ~3–6 GPU-years (modern B200-class GPU lattice siever)
- NFS@Home (10,000-core volunteer grid): ~1–2 months calendar time
- 1,000-core dedicated cluster: ~1–1.5 years
- ~50 modern GPUs (B200): ~1–2 months

Linear Algebra:

- ~150–400 CPU core-years on a tightly-coupled Infiniband cluster
- Matrix dimension ~300–500 million rows/columns
- Need 32–64 nodes, 256–512 GB RAM per node, Infiniband interconnect
- Calendar time: a few weeks to a few months
- This step is the real bottleneck and cannot be done on small cloud VMs

Storage & Power:

- Disk: ~5–15 TB (raw relations, before dedup)
- Cluster peak power: ~150–300 kW

2.5 Cloud Computing Cost Estimates

Using Cognition's market pricing ($0.08 / CPU core-hour, $3.50 / GPU-hour):

Route	Compute	Estimated Cost
Pure CPU cloud (sieving + LA)	~2,000 CPU core-yr ≈ 17.5 M CPU-hr	~$1.4 M
GPU sieving + dedicated LA cluster	~4 GPU-yr sieve + 3–6 mo LA cluster	~$150k–$500k
Academic / volunteer route	NFS@Home + XSEDE/TACC/SDSC allocation	$0 cash (donated compute)

Recommended: The academic/volunteer route — this is exactly how every prior SNFS record in the 320–384 digit range was actually done. NFS@Home has been sieving SNFS-340 class jobs and would be a natural host for SNFS-370. The linear algebra step requires an HPC allocation (XSEDE/TACC/SDSC), as M1061 used.

3. SNFS vs GNFS Comparison (for C284)

Approach	Work (L-function E)	Est. CPU core-yr	Verdict
GNFS on 284-digit C284	E ≈ 58.03	~69,000	INFEASIBLE — >20× largest GNFS ever (RSA
SNFS on M1229 (370-digit difficulty)	E ≈ 51.66	~1,500–2,500	FEASIBLE — within demonstrated capability
Raw work ratio GNFS/SNFS	exp(58.03−51.66) ≈ 588×	realised ~25–45×	SNFS wins by 1–2 orders of magnitude

3.1 Why SNFS Works on M1229

$M_{1229} = 2^{1229} - 1$ has the special form $a \cdot b^n + c$ with a=−1, b=2, n=1229, c=0 — all small. We pick a degree d and write 1229 = q·d + r, then use the polynomial $f(x) = x^d - 2^r$ (small coefficients!) with $m = 2^q$, so $f(m) = 2^{q \cdot d} - 2^r \equiv 0 \bmod (2^{1229} - 1)$ and hence mod C284.

Because f has tiny coefficients, the norms $b^d \cdot f(a/b)$ being sieved are roughly 2^{1229}, i.e. ~370-digit numbers — much smaller than the $\sim 10^{568}$ norms that GNFS would have to sieve for a generic 284-digit number. That is the entire source of the ~588× saving.

4. Software Tools for SNFS

Tool	License	Author / Maintainer	Role
CADO-NFS	LGPL (open source)	INRIA / LORIA: Thomé, Gaudry, Zimmermann, Bouvier,	Full NFS pipeline; state of the art; used for RSA-240/250/2
GGNFS (lasieve4)	GPL	Monico; pol5 by Kleinjung; lasieve4 by Franke, Kleinjung	Polynomial selection + lattice sieving; used by NFS@Hom
Msieve	Public domain	Jason Papadopoulos	Final-processing (linear algebra) + small-number polysel
YAFU	Public domain	Ben Buhrow	Modern alternative full pipeline
GMP-ECM	GPL	Zimmermann, Kruppa	ECM preprocessing (mandatory first step)
Kleinjung's lattice siever	research	Thorsten Kleinjung (Bonn/EPFL)	Fastest lattice siever; underlies GGNFS
NFS@Home	BOINC	Greg Childers, CSU Fullerton	Volunteer sieving grid; currently SNFS-340 jobs

Recommended stack for M_{1229}: CADO-NFS (full pipeline, SNFS mode) + GMP-ECM (final sanity check) + optionally NFS@Home for sieving distribution. For LA, either CADO-NFS's bundled block-Wiedemann on an Infiniband cluster, or Msieve's final-processing.

5. Why ECM Cannot Replace SNFS Here

5.1 The ECM Plateau

The all-time ECM record for largest factor found is 83 digits (Ryan Propper, September 2013). ECM has not found a larger factor in the past 13 years — it has plateaued.

Date	Largest Factor	Finder
Aug 2006	67 digits	B. Dodson
Dec 2009	68 digits	yoyo@home / M. Thompson
Mar 2010	73 digits	Bos, Kleinjung, Lenstra, Montgomery
Aug 2012	75 digits	Sam Wagstaff
Aug 2012	79 digits	Sam Wagstaff
Sep 2013	83 digits (CURRENT RECORD)	Ryan Propper

5.2 C284's Current ECM State

Per OEIS (the authoritative list of sequences needing factors): C284 of $2^{1229}-1$ has been ECM-tested to t65 (a 65-digit factor would have been found with high probability). The Cunningham Project convention further notes that base-2 Cunningham cofactors are typically ECM'd to t70. No factor below ~65–70 digits exists in C284.

5.3 ECM Effort to Find Larger Factors

ECM effort to find a d-digit factor scales as $L_p[1/2, \sqrt{2}] \approx \exp(\sqrt{(2 \cdot d \cdot \ln 10)})$ curve-equivalents. Relative to the 83-digit record:

Factor size searched	Relative effort vs. 83-digit record
83 digits (record)	1×
90 digits	~3×
100 digits	~7×
120 digits	~45×
142 digits (half of C284)	~460×
160 digits	~3,000×

Critical: Even a successful ECM hit on, say, a 75–90-digit factor would leave a ~195–210-digit cofactor that still needs SNFS or another large ECM find. ECM cannot replace SNFS for C284 — at best it might shrink the problem.

6. Bottom-Line Recommendation

6.1 Is SNFS Feasible for C284 of M1229?

Yes — clearly and unambiguously feasible. It is easier than the current SNFS world record (M_{1277}, SNFS-384, Sept 2026) and only ~1.7× the 2014 M_{1199} (SNFS-361) effort, which was completed more than a decade ago on 2.2 GHz Opterons.

6.2 What Would It Take?

Step	Effort	Calendar time	Hardware
1. Preprocessing (ECM)	Already done to t65/t70	0	—
2. Polynomial selection	A few days of compute	~1 week	1–2 workstations
3. Sieving	~1,000–1,500 CPU-yr (~3–6 GPU-yr)	~3 months	NFS@Home (10k cores) OR ~50 GPUs
4. Linear algebra	~150–400 CPU-yr	~weeks to months	32–64-node Infiniband cluster, 256–512 GB RAM/n
5. Storage	~5–15 TB scratch	—	—
6. Total calendar	—	~6–12 months	—

6.3 Cost Options

Route	Cash Cost	Notes
Volunteer / academic (recommended)	~$0	NFS@Home + HPC allocation; how every prior SNFS record was done
Commercial cloud (GPU route)	~$150k–$500k	Comparable to Cognition's $400k RSA-260 run
Commercial cloud (CPU-only)	~$1–2M	Not recommended — GPUs far more cost-effective now

6.4 Should ECM or GNFS Be Tried Instead?

ECM: No — C284 has already been ECM'd to t65/t70, the all-time ECM record is 83 digits (unchanged for 13 years), and even a heroic ECM find would only shrink the problem, not solve it. Further ECM effort is essentially wasted and should be spent on sieving instead.

GNFS: No — ~69,000 CPU core-years for the 284-digit cofactor is >25× more than SNFS and >20× the largest GNFS ever done (RSA-260). GNFS on a 284-digit general number is beyond current world capability; SNFS via the $2^{1229}-1$ form is the only viable path.

6.5 Final Verdict

Factoring C284 of M_{1229} is a routine Cunningham-Project-scale SNFS job. It is harder than what NFS@Home is currently running (~SNFS-340) but easier than the just-set M_{1277} record (SNFS-384). A well-organized effort could complete it within ~1 year using volunteer sieving plus an HPC LA allocation, at essentially zero cash cost. The result would be a high-quality SNFS factorization but would NOT set a new SNFS record (M_{1277} already holds that). It would be a worthy Cunningham-project contribution and a textbook demonstration of SNFS at the 370-digit difficulty level.

7. References

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· CADO-NFS homepage: https://cado-nfs.gitlabpages.inria.fr
· GMP-ECM: https://members.loria.fr (Zimmermann, Kruppa)